

# Natural genetic variation underlying the negative effect of elevated CO<sub>2</sub> on ionome composition in *Arabidopsis thaliana*

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## Reviewed Preprint

Revised by authors after peer review.

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**Version of Record published**  
May 23, 2024

**Reviewed preprint version 2**  
May 7, 2024 (this version)

**Reviewed preprint version 1**  
November 30, 2023

**Sent for peer review**  
July 5, 2023

**Posted to preprint server**  
June 7, 2023

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## Abstract

The elevation of atmospheric CO<sub>2</sub> leads to a decline in the plant mineral content, which might pose a significant threat to food security in the coming decades. To date, very few genes have been identified as having a role in the negative effect of elevated CO<sub>2</sub> on plant mineral composition. Yet, several studies have shown a certain degree of diversity in the ionome's response to elevated CO<sub>2</sub>, associated with genotypic variation. This suggests the existence of genetic factors controlling the effect of CO<sub>2</sub> on ionome composition. However, no large-scale studies have been carried out to date to explore the genetic diversity of the ionome responses to elevated CO<sub>2</sub>. Here, we used six hundred *Arabidopsis thaliana* accessions, representing geographical distributions ranging from worldwide to regional and local environments, to analyze the natural genetic variation underlying the negative effect of elevated CO<sub>2</sub> on the ionome composition in plants. We show that the growth under elevated CO<sub>2</sub> leads to a global and important decrease of the ionome content whatever the geographic distribution of the population. We also observed a high range of genetic diversity in the response of the ionome composition to elevated CO<sub>2</sub>, and we identified sub-populations, showing effects on their ionome ranging from the most pronounced to resilience or even to a benefit in response to elevated CO<sub>2</sub>. Using genome-wide association mapping on the response of each mineral element to elevated CO<sub>2</sub> or on integrative traits, we identified a large set of QTLs and genes associated with the ionome response to elevated CO<sub>2</sub>. Finally, we demonstrate that the function of one of these genes is associated to the negative effect of elevated CO<sub>2</sub> on the plant mineral composition. This resource will contribute to understand the genetic mechanisms underlying the negative effect of elevated CO<sub>2</sub> on plant mineral nutrition, and could help towards the development of crops adapted to a high-CO<sub>2</sub> world.

### eLife assessment

This paper provides **useful** information about how the ionome of *Arabidopsis thaliana* adapts to very high CO<sub>2</sub>-levels, backed up by **solid** evidence and carefully designed studies. The work will be of interest to anyone studying natural genetic variation as well as the response of plants to altered CO<sub>2</sub> levels in the atmosphere.

<https://doi.org/10.7554/eLife.90170.2.sa2>

## Introduction

The elevation of atmospheric CO<sub>2</sub> concentration leads to a decline in the mineral composition of C3 plants (Gojon et al., 2023 [↗](#)). The negative effect of elevated CO<sub>2</sub> on plant mineral composition has been observed worldwide, and alters the content of nutrients that are essential for human nutrition, such as nitrogen (N) and proteins, iron (Fe) or zinc (Zn) (Loladze, 2014 [↗](#)). Therefore, the rise in atmospheric CO<sub>2</sub> poses a significant threat to food security in the coming decades. Indeed, several modeling approaches predict a decrease in plant-based nutrient availability due to the negative effect of elevated CO<sub>2</sub> on the mineral status of plants, leading to an additional risk of nutritional deficiency for hundreds of millions of people (Medek et al., 2017 [↗](#); Ebi and Loladze, 2019 [↗](#)). However, the reasons why elevated CO<sub>2</sub> leads to the degradation of plant mineral composition are far from being well understood. To date, only a few genes with a potential regulatory role on this effect have been identified (Gao et al., 2019 [↗](#); Umnajkitikorn et al., 2020 [↗](#); Yang et al., 2020 [↗](#); Bouain et al., 2022 [↗](#); Sun et al., 2022 [↗](#); Cassan et al., 2023 [↗](#)). Despite their limited number, these studies show that the detrimental effect of high CO<sub>2</sub> on the plant mineral status has genetic bases. In addition to this, several reports suggest that exploring the natural genetic variability of plants represents a major opportunity to understand the mechanisms by which high CO<sub>2</sub> leads to a decline in plant mineral composition (Myers et al., 2014 [↗](#); Zhu et al., 2018 [↗](#); Marcos-Barbero et al., 2021 [↗](#)). Indeed, a significant diversity in the response of mineral composition to high CO<sub>2</sub> has been observed in several plant species. For protein and therefore N content, as well as for Fe or Zn content, substantial variations have been observed between small panels of genotypes from different species (Myers et al., 2014 [↗](#); Zhu et al., 2018 [↗](#); Marcos-Barbero et al., 2021 [↗](#)). This implies the presence of a genetic diversity reservoir, which can facilitate the understanding of the ionome's response to high CO<sub>2</sub> and subsequently provide an opportunity to alleviate this negative impact. However, in order to identify the genetic determinants of this negative response of the ionome to high CO<sub>2</sub>, large-scale approaches are necessary, but are still lacking for the moment. The objective of this work was to fill the aforementioned knowledge gap by using a large collection of natural genotypes of the model plant *Arabidopsis thaliana*. This allowed to explore in depth the natural variation of the ionome response to elevated CO<sub>2</sub>, and to generate a resource of phenotypic data that can be used in association genetics approaches. To this end, we used several hundreds of accessions from different geographic scales of *A. thaliana*, and analyzed their leaf mineral composition under contrasted conditions of CO<sub>2</sub> concentration. This allowed us to extract the general trends in the leaf ionome response to high CO<sub>2</sub>, and to identify a large set of genes associated with the variation in the mineral composition of plants in response to high CO<sub>2</sub>. By combining this information with genome expression data under elevated CO<sub>2</sub>, we end up by functionally validating one of these genes for its importance in the reduction of Zn content under elevated CO<sub>2</sub>. Our results open the way for a better understanding of the genetic and molecular mechanisms involved in the regulation of plant mineral nutrition by the elevation of atmospheric CO<sub>2</sub>.

## Results

In order to explore the natural variation and identify its underlying genetic basis associated with the negative effect of elevated CO<sub>2</sub> on plant ionome, we used three populations of *A. thaliana* representing different geographic scales (i.e., the worldwide REGMAP population, the LANGUEDOC regional population and the local TOU-A population from east of France) and displaying different levels of genetic diversity (**Fig. 1A**). These populations were grown under ambient or elevated CO<sub>2</sub>, and we measured in each accession the composition of their ionome in rosettes, including C, N, Na, Fe, Mg, Mn, Zn and Cu content.

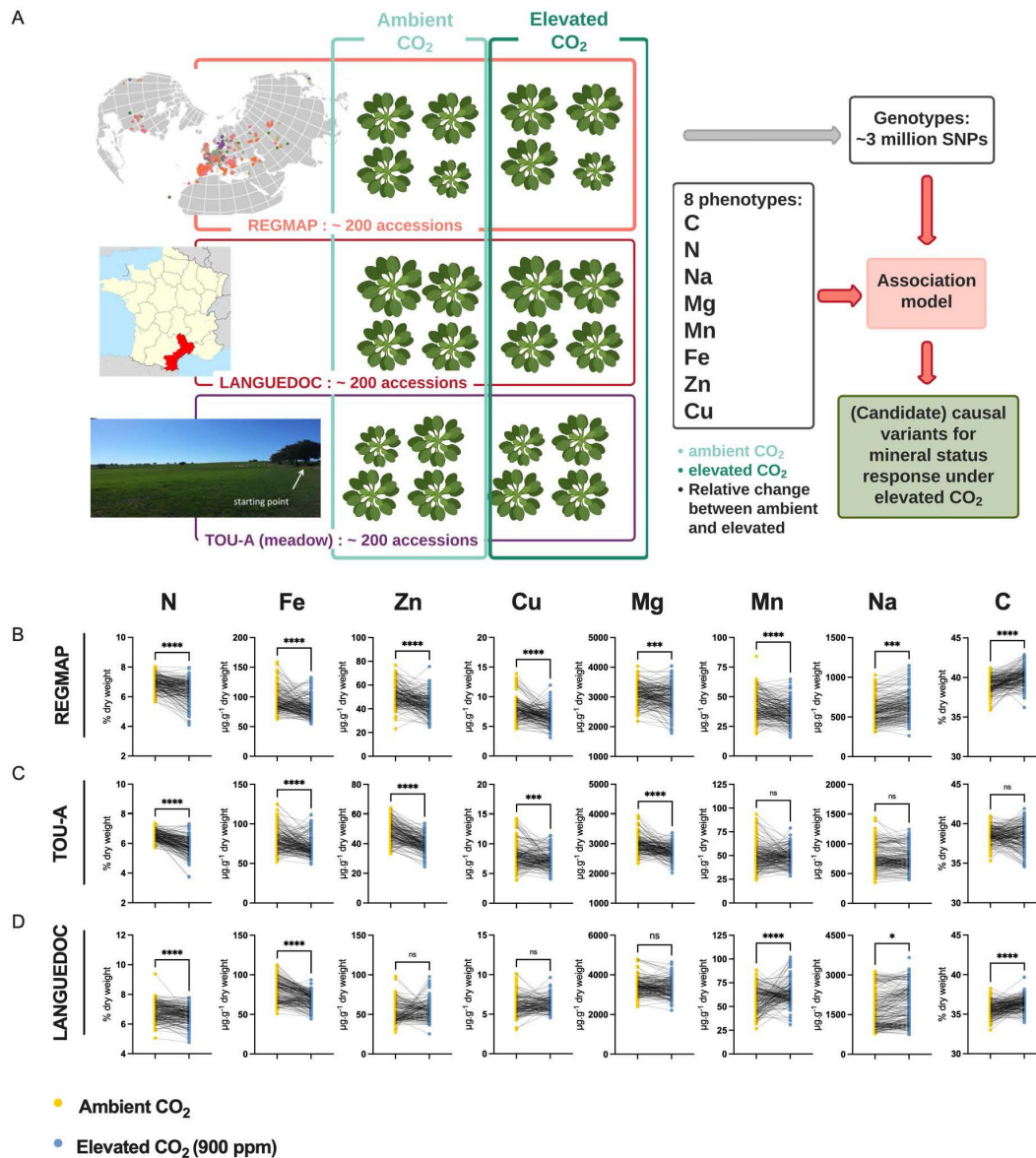
### Elevated CO<sub>2</sub> globally decreases ionome content at the population level, whatever the geographic scale

In the three *A. thaliana* populations, we observed a global and important decrease of the ionome content when plants were grown under elevated CO<sub>2</sub> as compared to ambient CO<sub>2</sub>. This was particularly the case for N and Fe, for which the decrease in content was very robust and important in each of the population analyzed (**Fig. 1B-D**). Zn, Cu and Mg content were also negatively affected to a significant extent by the growth under elevated CO<sub>2</sub> in the REGMAP and in the TOU-A populations (**Fig. 1B, C**), although not significantly in the LANGUEDOC population (**Fig. 1D**). More variability for the effect of elevated CO<sub>2</sub> was observed on Mn and Na content, which were decreased in the REGMAP population, but not significantly changed in the TOU-A and LANGUEDOC populations, respectively. In parallel, the C content of these populations increased under elevated CO<sub>2</sub>, by very significant factors for the REGMAP and the LANGUEDOC populations. Altogether, these observations demonstrate that elevated CO<sub>2</sub> has on average a negative impact on the mineral content of natural genotypes of *A. thaliana* at the population-scale, whatever their geographic distribution.

### The ionome of *Arabidopsis thaliana* natural accessions displays a high range of phenotypic diversity in response to elevated CO<sub>2</sub>

To explore the effect of elevated CO<sub>2</sub> in each accession, we calculated the relative change in nutrient composition of *A. thaliana* accessions from the three populations in response to elevated CO<sub>2</sub>. In agreement with the results previously mentioned, we observed that the median relative change of most nutrient content at the population-level was negatively affected by elevated CO<sub>2</sub> (**Fig. 2**). But the most striking observation was the genetic diversity of ionome response observed in these populations. Indeed, while most the natural accessions were negatively affected by elevated CO<sub>2</sub> (with a negative relative ratio of their nutrient content between ambient and elevated CO<sub>2</sub>), a considerable number of accessions were rather not affected by elevated CO<sub>2</sub>, or even positively affected, therefore showing an improved nutrient composition under elevated CO<sub>2</sub>. For macronutrients like N, the relative change of concentration between ambient and elevated CO<sub>2</sub> varied from 20% to -50%, and for micronutrients like Cu, Fe or Zn, the relative change of concentration between ambient and elevated CO<sub>2</sub> varied from 100% to -60% (**Fig. 2**). In addition, some differences among nutrients were observed between populations. For instance, a smaller dispersion of Fe relative change in the LANGUEDOC population, against a higher distribution of Mn relative change.

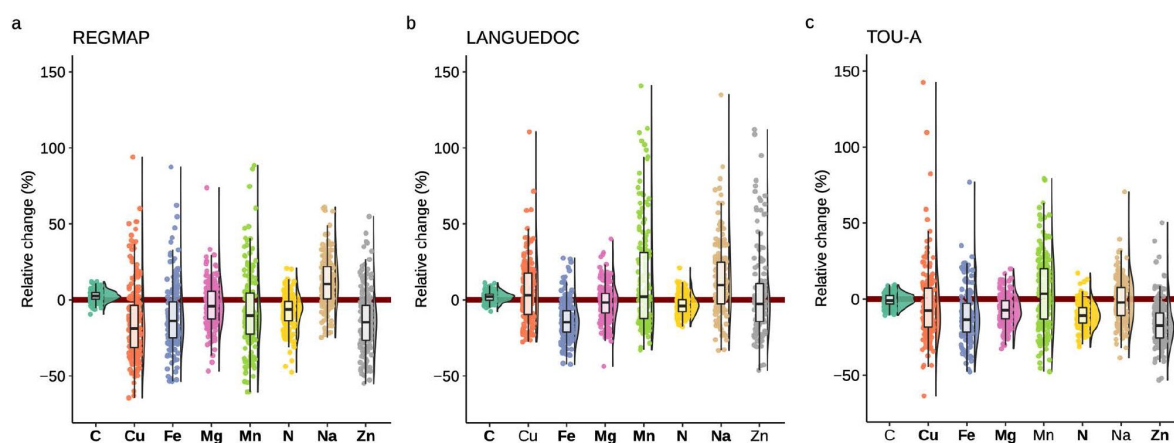
In order to explore the behavior of the different elements in response to elevated CO<sub>2</sub> and to observe the structure of phenotypic variation, we performed a principal component analysis (PCA) of the relative change in the 8 elements for the accessions from the three populations. The accessions from all populations seem to have globally similar responses to elevated CO<sub>2</sub>, as



**Figure 1**

### Elevated CO<sub>2</sub> negatively impacts the ionome content at the population-scale level in *Arabidopsis thaliana*.

A. Representation of the experimental design used in this study. The content of eight mineral elements was assessed for around 600 *Arabidopsis thaliana* accessions coming from the REGMAP (B), LANGUEDOC (C) and TOU-A (D) populations. Each dot represents the value of the content of a mineral element for one accession (yellow: ambient CO<sub>2</sub> (aCO<sub>2</sub>, ~420 ppm), blue: elevated CO<sub>2</sub> (eCO<sub>2</sub>, 900 ppm). N (% of dry weight), Fe ( $\mu\text{g g}^{-1}$  dry weight), Zn ( $\mu\text{g g}^{-1}$  dry weight), Cu ( $\mu\text{g g}^{-1}$  dry weight), Mg ( $\mu\text{g g}^{-1}$  dry weight), Mn ( $\mu\text{g g}^{-1}$  dry weight), Na ( $\mu\text{g g}^{-1}$  dry weight), C (% of dry weight). Asterisks indicate significant differences (Paired Wilcoxon test; \*,  $P < 0.05$ ; \*\*,  $P < 0.005$ ; \*\*\*,  $P < 0.0005$ ). ns; not significant.



**Figure 2**

**Elevated CO<sub>2</sub> leads to high phenotypic diversity of ionome response in *Arabidopsis thaliana*.**

Distributions of the relative change (%) of the content of 8 mineral elements between elevated CO<sub>2</sub> and ambient CO<sub>2</sub>, in each population (A: REGMAP, B: LANGUEDOC, C: TOU-A). Each dot represents the value of the relative change of the content a mineral element for one accession. The name of the element appears in bold if the mean of the element in elevated CO<sub>2</sub> is significantly different from the mean of the element in ambient CO<sub>2</sub> (Paired wilcoxon test, significance threshold of 0.05).

suggested by the overlap of the three populations in the two first principal components (**Fig. 3A**). The first component of the PCA described a clear antagonistic trend between C content and the change of other mineral elements (**Fig. 3B**), suggesting that most of the variation between accessions in term of mineral response (almost 40%) could be driven by one or a few mechanisms resulting in an inverse variation between the whole ionome and C change (**Fig. 3B**). Interestingly, the second component, explaining almost 15% of the variation among accessions in term of mineral response, was mainly driven jointly by change in N and C concentration. Altogether, these results show that there is a marked and large variability among accessions in their mineral concentration in response to elevated CO<sub>2</sub>, illustrated by accessions negatively affected by elevated CO<sub>2</sub> and others positively affected by elevated CO<sub>2</sub>.

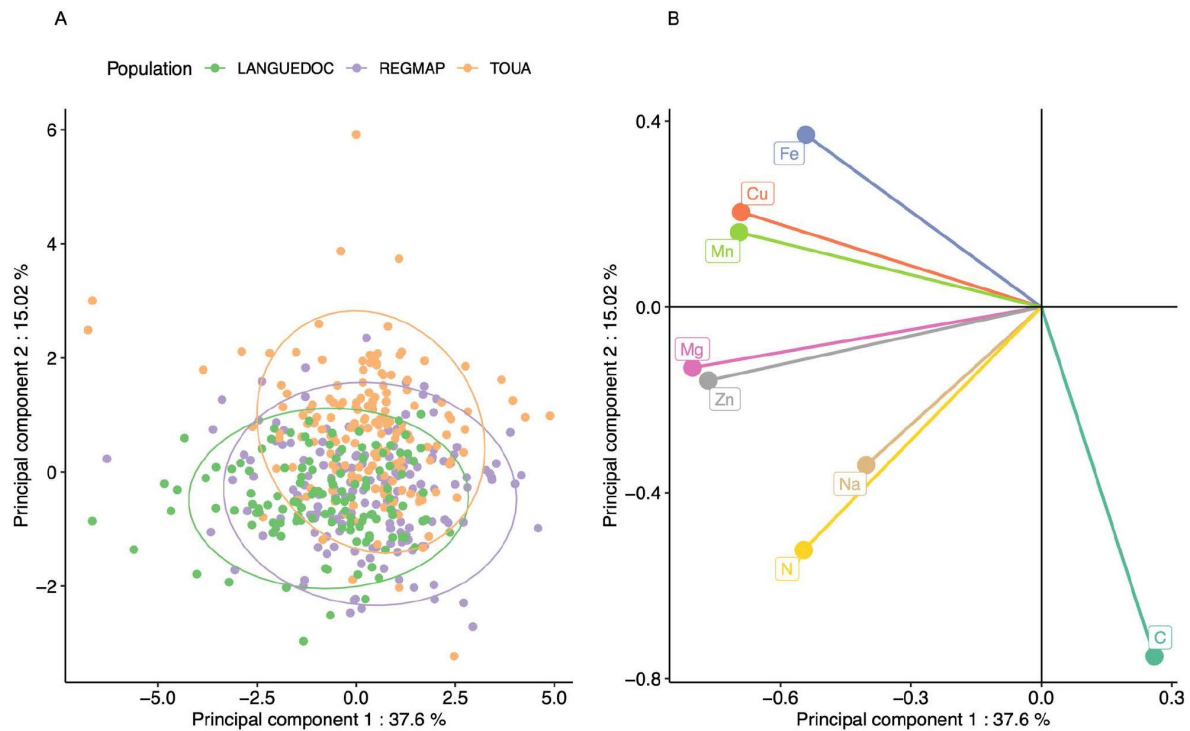
In order to explore specific behavior of sub-populations, we clustered the accessions from the REGMAP panel via a k-means approach. This multivariate clustering resulted in the partitioning of accessions in three groups (**Fig. 4** – Suppl. Table 1). Cluster 1 displayed the most negative pattern of ionome response to elevated CO<sub>2</sub>. Inversely, accessions included in Cluster 2 displayed a globally positive response, with the highest relative change for almost all mineral elements, except for C content. These accessions did not appear to be clustered geographically with respect to their collection origin in the REGMAP panel (Suppl. Fig. 1), which is in line with the high genetic diversity of response to elevated CO<sub>2</sub> observed at smaller geographical scales (**Fig. 2**). Finally, Cluster 3 displayed a resilient pattern, with accessions showing a globally attenuated response to elevated CO<sub>2</sub>. Interestingly, the large phenotypic diversity of the ionome observed in the three populations in response to high CO<sub>2</sub>, as well as the presence of contrasted subpopulations in the REGMAP panel, suggests the presence of genetic determinants associated with this response.

## Genetic architecture of the ionome response to elevated CO<sub>2</sub>, and identification of genetic determinants

We ran Genome-Wide Association (GWA) mapping to describe the genetic architecture of the ionome response to elevated CO<sub>2</sub>, and to fine-map candidate genes underlying the detected quantitative trait loci (QTLs). We focused here on the phenotypic data collected on the REGMAP population, and used the sequencing data available for this population (Arouisse et al., 2020). We included in this analysis the level of each mineral under ambient and under elevated CO<sub>2</sub>, as well as the relative change between ambient and elevated CO<sub>2</sub> for each element. We also included a trait corresponding for each accession to the coordinate on the first and on the second PCA axes (PCA1 and PCA2) explaining collectively more than 50% of ionic variation (**Fig. 3**). Therefore, these values correspond to traits driving and summarizing a large part of the ionome variation under elevated CO<sub>2</sub>. This resulted as a whole in running GWA mapping on 30 different single-trait GWAS. The overall approach was first validated by observing expected results for traits phenotyped under ambient CO<sub>2</sub>. For instance, we observed a very strong peak for the Na content at the locus of the *HKT1* gene (Suppl. Fig. 2A), which is known to be involved in the natural genetic variation of Na content in *Arabidopsis thaliana* (Baxter et al., 2010), or a strong peak for the N content at the locus of the *NIA1* gene (Suppl. Fig. 2B), encoding for an isoform of the nitrate reductase required for the first step of nitrate reduction and associated with natural genetic variation of N content in *A. thaliana* (North et al., 2009).

GWA mapping revealed a polygenic architecture for each phenotypic trait, although its complexity largely differs among traits. For instance, very few and neat peaks of association were detected Na and Mn content under elevated CO<sub>2</sub>, or of Fe and Cu relative change between ambient and elevated CO<sub>2</sub> (**Fig. 5A**, Suppl. Fig. 3). On the other hand, a more complex genetic architecture with the detection of a large number of QTLs was observed for traits related to N or C content (**Fig. 5A**, Suppl. Fig. 3). For each of the traits that have been analyzed under elevated CO<sub>2</sub> or corresponding to the relative change of their content between ambient and elevated CO<sub>2</sub>, we isolated the 50 SNPs with the most significant p-value, hereafter named top SNPs (**Fig. 5A**, Suppl.

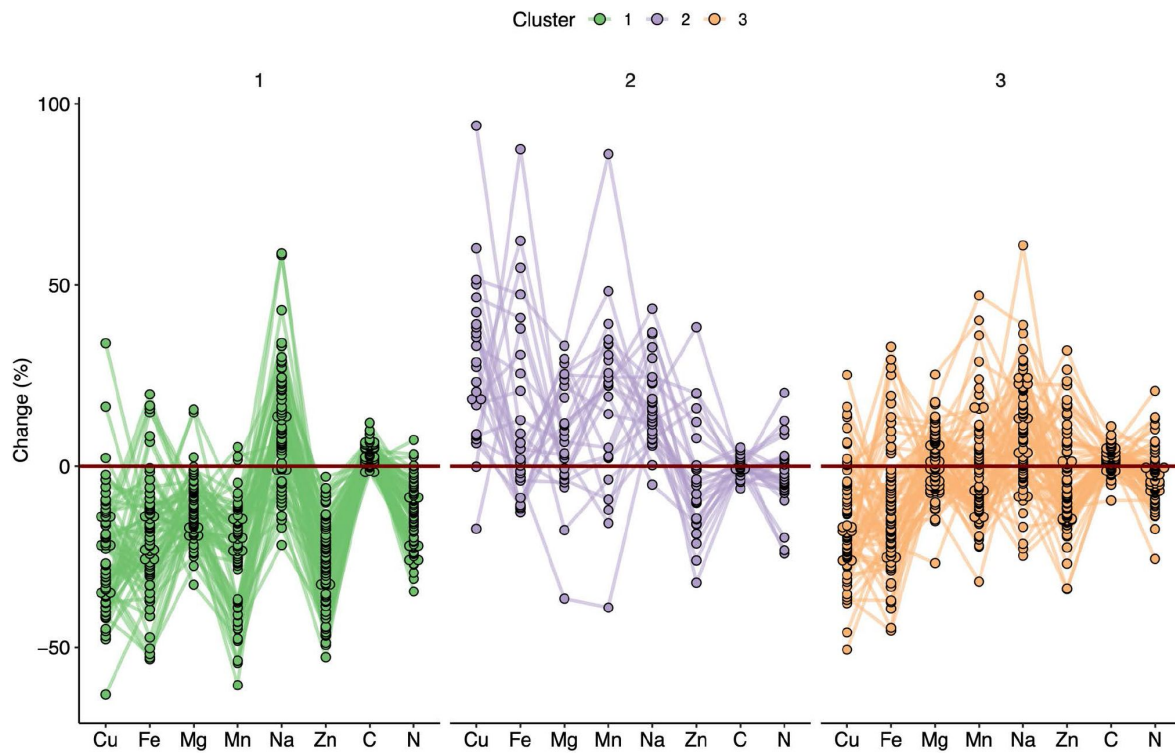




**Figure 3**

**Elevated CO<sub>2</sub> results in a general pattern of ionome variation common to most accessions constituting natural populations of *Arabidopsis thaliana*.**

Principal Component Analysis (PCA) was performed using the variation of each element in response to elevated CO<sub>2</sub>. A. Natural accessions were positioned on the PCA and colored based on population. B. Contribution of each element to the PCA axis.



**Figure 4**

**Variation in the response of the ionome to elevated CO<sub>2</sub> identifies contrasting subpopulations inside the REGMAP panel.**

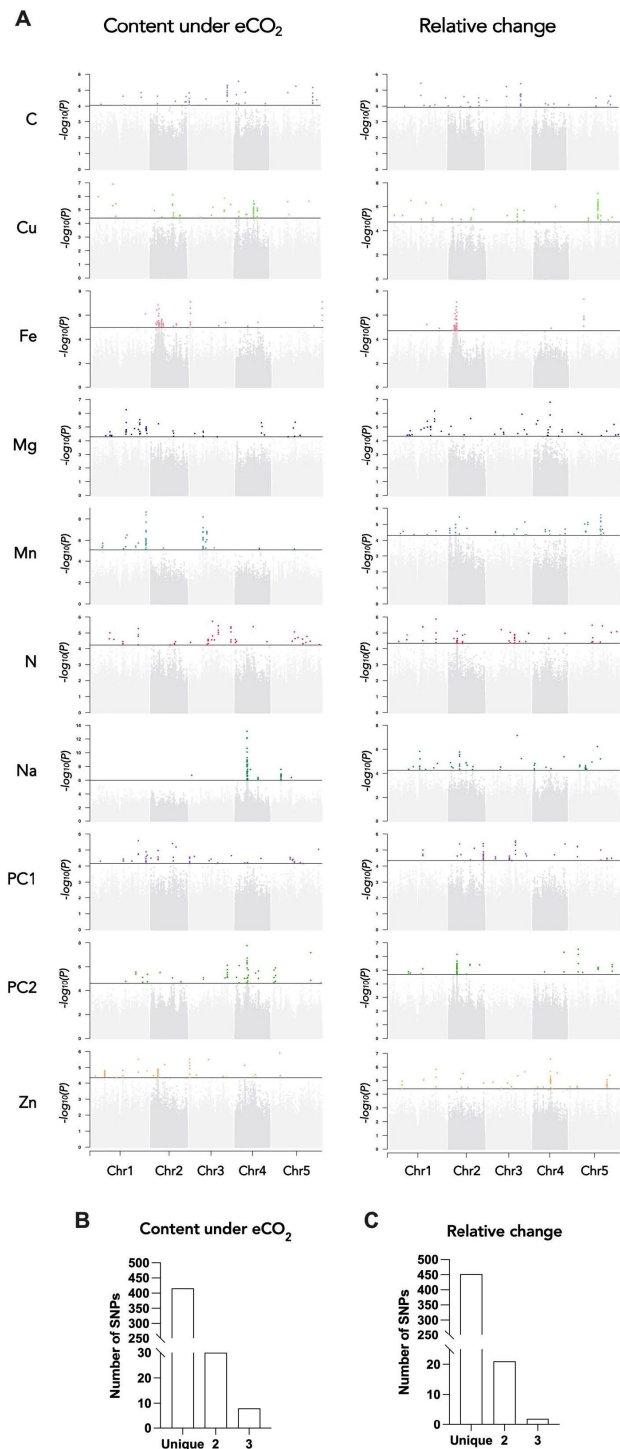
K-means clustering was performed in the REGMAP accessions to identify different subpopulations. Each accession is represented by a dot, connected by a line between each element. Cluster 1: 65 accessions. Cluster 2: 25 accessions. Cluster 3: 69 accessions.



Tables 2 and 3). In order to identify the overlap between the genetic architecture of each trait, we looked whether some of the top SNPs were shared among traits. While the large majority of SNPs were specific to one trait, 30 and 21 SNPs were shared between two traits for the content under elevated CO<sub>2</sub> or for the relative change between ambient and elevated CO<sub>2</sub>, respectively (Fig. 5B and C, Supplemental Tables 2 and 3). In addition, 8 and 2 SNPs were shared between three traits for the content under elevated CO<sub>2</sub> or for the relative change between ambient and elevated CO<sub>2</sub>, respectively (Fig. 5B and C, Supplemental Tables 2 and 3). Most of the shared SNPs were associated with micronutrients (Fe, Mn, Zn and Mg content) and with N and/or with the first component of the PCA axis. An interesting QTL located on chromosome 1 was notably associated with 6 traits, displaying SNPs shared between Mn, Zn and N relative change and SNPs shared between Mn, N and PC1 content under elevated CO<sub>2</sub> (Fig. 5A, Suppl. Tables 2 and 3). Another QTL located on chromosome 3 encompasses SNPs shared between Fe, Zn and PC1 content under elevated CO<sub>2</sub> (Fig. 5A, Suppl. Table 2).

We next identified for each trait a list of the genes located at  $\pm 25$  kb from the top 50 SNPs, which corresponds to the rough estimate of the decay of linkage disequilibrium identified in *A. thaliana* at the worldwide scale (Kim et al., 2007). This resulted in a list of genes for each element, ranging from 154 to 422 genes depending on the element (Suppl. Tables 2 and 3). Among others, several genes associated with top 50 SNPs were identified as obvious candidates of the effect of elevated CO<sub>2</sub> on plant nutrition and ionome content. This was the case of *ZINC INDUCED FACILITATOR 1* (*ZIF1*, *AT5G13740*) and *ZIF-LIKE1* (*AT5G13750*), linked with SNPs identified for Zn content under elevated CO<sub>2</sub>, and involved Zn sequestration mechanisms (Lee et al., 2021). We also noticed the link between SNPs identified for Zn relative change and *TIP2;2* (*AT4G17340*), known to be involved in Zn root-to-shoot translocation (Wang et al., 2022). Concerning N relative change, some of the top 50 SNPs were linked to *ASN1* (*AT3G47340*), which is an actor of N status and remobilization (Lam et al., 2003; Gaufichon et al., 2017). Some of the top 50 SNPs identified for Fe relative change were linked to *MCO2* (*AT5G21100*) and *MCO3* (*AT5G21105*) genes, which have been recently characterized as actors of the regulation of Fe homeostasis (Brun et al., 2022). Finally, it is interesting to note that the QTL located on chromosome 3 mentioned above displaying significant shared SNPs identified for Fe, Zn and PC1 content under elevated CO<sub>2</sub> was associated among other genes with *ISU2* (*AT3G01020*), coding for one of the Fe-S clusters in *Arabidopsis thaliana*, which are known to be essential for photosynthesis and metabolism (Balk and Schaedler, 2014). Altogether, this demonstrated that genes identified through this approach represent a large and valuable reservoir of candidates to study and to counteract the effect of elevated CO<sub>2</sub> on plant nutrition and ionome content.

To analyze how these genes identified by GWA mapping are regulated by elevated CO<sub>2</sub>, we performed RNA-seq from shoots and roots grown under ambient and elevated CO<sub>2</sub>. Differentially expressed genes (DEG) associated to the effect of elevated CO<sub>2</sub> were identified from shoots and roots (Suppl. Table 4). We compared the list of shoots or roots elevated CO<sub>2</sub>-DEG with the list of genes identified by GWA mapping for each element, which resulted in a list of 182 genes identified by GWA mapping and differentially regulated by elevated CO<sub>2</sub> in shoot or in roots (Suppl. Table 5), making them relevant candidates to be involved in the response of the mineral composition of plants to elevated CO<sub>2</sub>. Most of these genes were deregulated by elevated CO<sub>2</sub> in shoot (Fig. 6A, B). In shoot or in roots, these genes mainly showed an association with C-, Mg- or Zn-related traits (Fig. 6A, B). Several of these genes, identified by GWA mapping and whose expression is deregulated in response to high CO<sub>2</sub>, were known for their role in nutrient homeostasis. This was the case for the *ASN1* and *DUR3* genes, encoding an asparagine synthase and a urea transporter involved in N metabolism and remobilization, both associated here with a N-related peak of association, and whose expression is modulated by high CO<sub>2</sub> in leaves (Fig. 6C). We also observed in the leaves an interesting profile for several genes related to C metabolism and photosynthesis. This was the case for the *BGP3* gene, involved in chloroplast development, or for the carbonic anhydrase *CA1*, both showing a decreased expression in response to high CO<sub>2</sub> and both associated with a peak in C-related GWA mapping under elevated CO<sub>2</sub> (Fig. 6C). In roots,



**Figure 5**

**Genetic architecture of the response of the ionome to elevated CO<sub>2</sub> in the REGMAP panel of *Arabidopsis thaliana*.**

A. Manhattan plots for the content of eight mineral elements under elevated CO<sub>2</sub>, or for the relative change of the content of mineral elements between elevated CO<sub>2</sub> and ambient CO<sub>2</sub>. For each Manhattan plot, SNPs with the 50 most significant P-value, located above the horizontal line, are colored. Bar plots showing the number of SNPs identified by GWAs for traits under elevated CO<sub>2</sub> (B) or for the relative change of the content of mineral elements between elevated CO<sub>2</sub> and ambient CO<sub>2</sub> (C) that are unique to one element or shared between 2 or 3 traits.

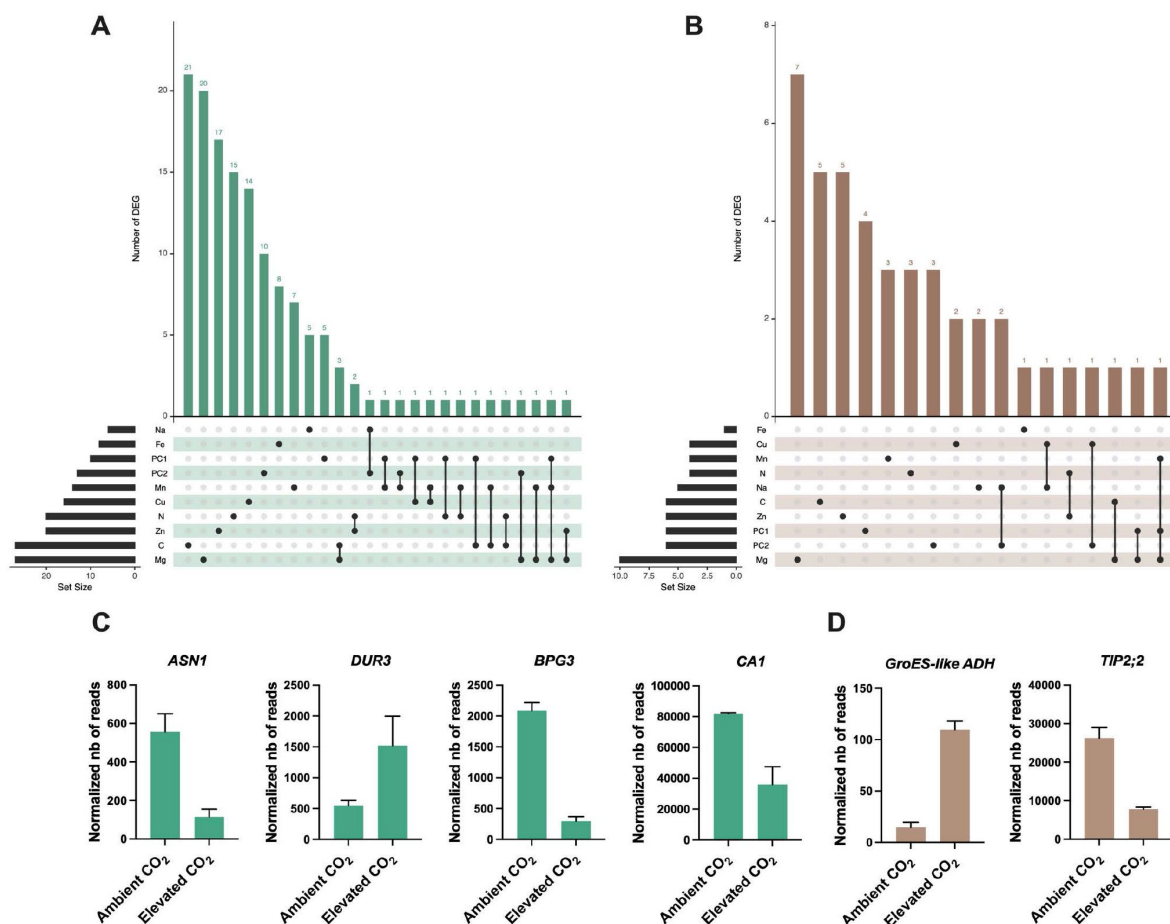
the gene most deregulated in response to high CO<sub>2</sub> was *AT1G64710*, encoding a GroES-type alcohol dehydrogenase, which interestingly is also deregulated in leaves (Fig. 6D). We also observed in the roots that the expression of the *TIP2;2* gene, associated here with a peak detected for Zn relative change GWA mapping, was deregulated in response to elevated CO<sub>2</sub> (Fig. 6D).

To go further, we selected one of the association peaks identified by GWA mapping, and sought to functionally validate the importance of this QTL in response to elevated CO<sub>2</sub>, in order to demonstrate the value of our data set and GWA mapping analyses. To do so, we selected an association peak located on chromosome 4 and associated with Zn relative change (Fig. 7A). More precisely, this association peak displayed the SNPs with the most significant *p*-values and more largely three SNPs that fell into the top 10 SNPs of the trait corresponding to the Zn relative change between ambient and elevated CO<sub>2</sub>. The SNPs corresponding to the alternative alleles were associated to an increase of Zn content under elevated CO<sub>2</sub> (Fig. 7B). These SNPs are located very close to the *TIP2;2* (*AT4G17340*) gene, which has been recently characterized as an actor of Zn root-to-shoot translocation (Wang et al., 2022). We thus selected a set of accessions from haplotype 0 (reduced Zn content under elevated CO<sub>2</sub>) or haplotype 1 (increased Zn content under elevated CO<sub>2</sub>), and analyzed *TIP2;2* expression in the roots under ambient and elevated CO<sub>2</sub>. This analysis revealed a haplotype-specific difference in *TIP2;2* expression: under ambient and elevated CO<sub>2</sub>, accessions from haplotype 1 (correlated with a higher Zn content in the shoot (Fig. 7B)) show a reduced *TIP2;2* expression in the roots compared to those from haplotype 0 (Fig. 7C & D), with a reduction being more pronounced under elevated CO<sub>2</sub>. To test the effect of *TIP2;2* expression of Zn content under elevated CO<sub>2</sub>, we used the *tip2;2-1* knock-out mutant and compared its Zn content under ambient and elevated CO<sub>2</sub> to this of the WT. We observed that the *tip2;2-1* mutant line did not present any decrease in Zn shoot content in response to elevated CO<sub>2</sub>, in opposition to what is observed for the WT (Fig. 7E). In the *tip2;2-1* mutant, Zn content under elevated CO<sub>2</sub> was even slightly higher than under ambient CO<sub>2</sub>. Altogether, these results demonstrated that these data sets generated in this study and the associated analyses are a valuable resource to identify genes and associated mechanisms involved in the effect of elevated CO<sub>2</sub> on the mineral composition of plants.

## Discussion

### The natural variation of ionome response to elevated CO<sub>2</sub> in *Arabidopsis thaliana* displays a high degree of genetic variation

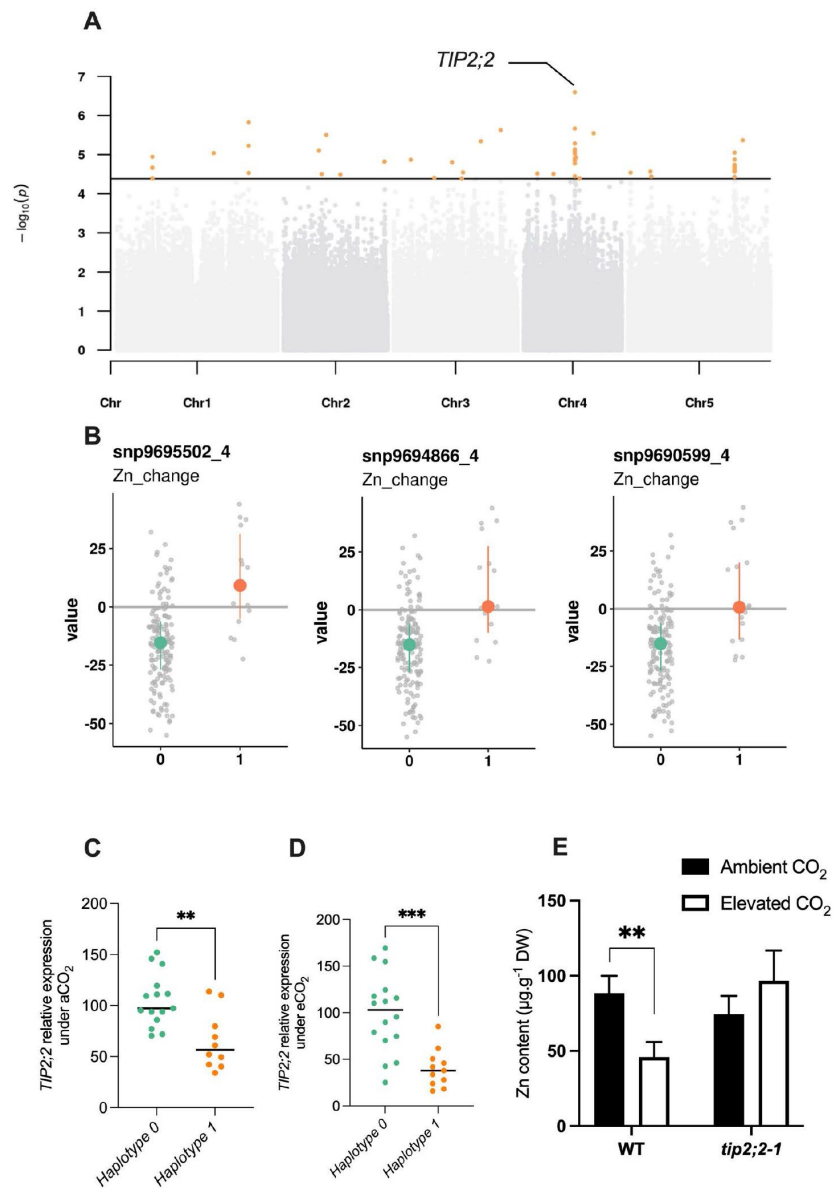
In the present work, we analyzed the diversity of leaf ionome response to elevated CO<sub>2</sub> present in the natural variation of *Arabidopsis thaliana*. In agreement with several other phenotypic traits related to phenology and disease resistance (Brachi et al., 2013; Huard-Chauveau et al., 2013; Roux and Frachon, 2022), we observed a wide range of responses at complementary geographical scales, from accessions with an ionome negatively affected by high CO<sub>2</sub> to accessions with an ionome benefiting from high CO<sub>2</sub>. This confirms for the first time on large and complementary sets of natural genotypes what has been observed by meta-analysis on isolated groups of plants worldwide (Loladze, 2014; Myers et al., 2014). The global analysis of the distribution of each mineral element studied suggests firstly a trend where the whole ionome would evolve in a unified manner in response to high CO<sub>2</sub>, and in an opposite manner to C. This is in line with a number of studies that have proposed that the accumulation of carbohydrates due to the stimulation of photosynthesis by high CO<sub>2</sub> would be the cause of the decrease in plant mineral composition (Ainsworth and Long, 2005; Thompson et al., 2017; Dusenge et al., 2019; Tausz-Posch et al., 2020). However, the reading of the genetic architecture performed here by a genome-wide association genetics approach suggests that the majority of the genetic mechanisms underlying the negative effect of elevated CO<sub>2</sub> on the ionome are specific to each mineral element.



**Figure 6**

### Identification of genes detected by GWA mapping and differentially regulated by elevated CO<sub>2</sub>.

Intersection between elevated CO<sub>2</sub>-DEG in shoot (A) or root (B) and genes identified by GWA mapping. UpSet plots display the number of elevated CO<sub>2</sub>-DEG that are associated to a locus identified for the content or the relative change of one or several mineral elements under elevated CO<sub>2</sub>. Illustration of the pattern of elevated CO<sub>2</sub>-DEG in shoot (C) or root (D) also identified by GWA mapping.



**Figure 7**

### Natural variation of the *TIP2;2* gene is associated with improved responses of Zn content to elevated CO<sub>2</sub>.

**A.** Manhattan plot of the relative change of Zn content between elevated CO<sub>2</sub> and ambient CO<sub>2</sub> showing the presence of a peak closed to the *TIP2;2* locus. **B.** Comparison of haplotypes and their relative change of Zn content between elevated CO<sub>2</sub> and ambient CO<sub>2</sub>. Three SNPs located at the *TIP2;2* locus are associated to an improvement of Zn content under elevated CO<sub>2</sub> for accessions that possess them (haplotype 1) compared to the rest of the population (haplotype 0). **C, D.** Relative expression of *TIP2;2* in the roots under ambient (**C**) or elevated (**D**) CO<sub>2</sub> for accessions belonging to haplotype 0 or haplotype 1. Relative expression levels were calculated based on *UBQ10* as internal control. Horizontal black line represented the median of each group of haplotypes. \*\*\* $P < 0,001$ , \*\* $P < 0,01$ , unpaired Mann-Whitney test. **E.** Shoot Zn content under ambient or elevated CO<sub>2</sub> for WT (Columbia) and *tip2;2-1* mutant. Data are presented as the mean (with SD) of 5 and 6 biological repeats for the WT and *tip2;2-1*, respectively. \*\* $P < 0.01$ , unpaired Mann-Whitney test.

Some specific cases, such as the QTL detected on chromosome 1 and associated with the natural genetic variation of 6 traits among the 20 considered, will certainly deserve a more in-depth analysis.

By clustering globally distributed accessions according to their ionome sensitivity to high CO<sub>2</sub>, we were able to observe that the geographic origin of the accessions likely did not determine their response to CO<sub>2</sub>. This suggests that inherent genetic factors, more than those due to local adaption, direct the response of plants to elevated CO<sub>2</sub>. This seems consistent since the CO<sub>2</sub> elevation applied here to natural *Arabidopsis thaliana* variants does not correspond to any environment experienced by plants yet, at least for several tens of millions of years (Pearson and Palmer, 2000 [↗](#); Luthi et al., 2008 [↗](#)). In this context of brutal and highly impactful environmental change, the presence of cryptic genetic variation often explains the appearance of relatively rapid adaptive mechanisms (Pauls et al., 2013 [↗](#); Cortés and López-Hernández, 2021 [↗](#)). Although not formally tested here, it would be interesting to examine whether the variation in the ionome in response to elevated CO<sub>2</sub> shows evidence of cryptic variation. In any case, the presence of high phenotypic diversity in these natural populations of *A. thaliana* demonstrates very clearly the possibility of taking advantage of this genetic variation to understand and alleviate the negative response of plant mineral composition to high CO<sub>2</sub>.

## GWA mapping of ionome variation under elevated CO<sub>2</sub> identified a large number of genes to understand and mitigate the negative effect of high CO<sub>2</sub> on plant mineral composition

In order to understand the genetic mechanisms underlying the effect of high CO<sub>2</sub> on plant mineral composition, and to enable future breeding approaches, we adopted an association genetics approach. This led to the identification of a large number of candidate genes associated to the variation of nutrients under elevated CO<sub>2</sub>. Several genes in this list can easily attract attention. In particular, we can note the identification of *ASN1* and *DUR3* genes in two of the loci associated with N content variation under elevated CO<sub>2</sub>. *ASN1*, and to a lesser extent *DUR3*, play an important role in the remobilization and the reallocation of N within the plant, and their manipulation can lead to variation in N use efficiency (Lam et al., 2003 [↗](#); Bohner et al., 2015 [↗](#); Gaufichon et al., 2017 [↗](#)). This is interesting because for the moment, root N uptake and N assimilation seemed to be the key targets of the negative effect of high CO<sub>2</sub> on plant N content (Bloom et al., 2010 [↗](#); Cassan et al., 2023 [↗](#)), but these results suggest that remobilization of N may also be involved. We also identified the *CA1* gene, coding for a carbonic anhydrase, in the vicinity of a QTL associated with C variation under high CO<sub>2</sub>. *CA1* is involved in the regulation of stomatal opening by elevated CO<sub>2</sub> (Hu et al., 2015 [↗](#)), and the  $\beta$  carbonic anhydrase family of which *CA1* belongs is involved in the regulation of photosynthetic efficiency, although *CA1* shows no significant effect under standard conditions (Sharma et al.). It would be therefore interesting to assess the role of *CA1* natural genetic variation under elevated CO<sub>2</sub>. If *CA1* regulates the C variation of the ionome under elevated CO<sub>2</sub>, this could, according to our observations, significantly influence the global mineral composition of plants. Interestingly, the genes identified by GWA mapping in the ionome response to high CO<sub>2</sub>, including those mentioned above, showed substantial variation at the gene expression level. We ended this study with the functional validation of an association peak identified by GWA mapping for the relative change of Zn content between ambient and elevated CO<sub>2</sub>. Zn is an essential element for a large number of metabolic processes in humans, and Zn deficiency, found in up to one third of the world's population, leads to severe health problems. We demonstrated that *TIP2;2* gene expression varied in a haplotype-specific manner, in both ambient and elevated CO<sub>2</sub>. In parallel to this, we show that loss of *TIP2;2* expression using a knock-out mutant can abolish the Zn decrease observed under high CO<sub>2</sub>. It therefore seems that variation in *TIP* expression is associated with the effect that atmospheric CO<sub>2</sub> can have on zinc levels. A recent study demonstrated that *TIP2;2* was responsible for Zn retention in the roots (Wang et al., 2022 [↗](#)). It therefore seems consistent that natural accessions with the lowest expression levels of this gene



are those with the highest Zn content in aerial parts, due to low retention in their roots. This example illustrates the potential of the resource we have generated here towards the identification of genes involved in the variation of leaf ionome in response to rising CO<sub>2</sub>, and towards the characterization of the associated mechanisms. The understanding of these mechanisms represents a considerable challenge in view of the current rise in atmospheric CO<sub>2</sub>, which might be useful to the coming breeding programs for crops adapted to a forthcoming CO<sub>2</sub>-rich atmosphere (Shahzad and Rouached, 2022 [↗](#)). It must however be recalled that leaf ionome is not always correlated with seed ionome, in particular in *Arabidopsis thaliana* accessions (Campos et al., 2021 [↗](#)). Therefore, the effect of elevated CO<sub>2</sub> on seed ionome, certainly in major crops, must be the next target to analyze in order to pave the way for the development of nutritious crops adapted to elevated CO<sub>2</sub>.

## Methods

### Data and code availability

Data and R notebooks containing the analyses performed in this article can be found at <https://src.koda.cnrs.fr/groups/ipsim/sirene-team> [↗](#). RNA-seq data generated for this study are available at <https://www.ebi.ac.uk/biostudies/arrayexpress/studies> [↗](#) using the accession no E-MTAB-13661.

### Plant Material

A subset of the REGMAP panel, the LANGUEDOC panel and the TOU-A panel were used in this study. These populations were previously described here (Horton et al., 2012 [↗](#); Brachi et al., 2013 [↗](#); Frachon et al., 2017 [↗](#)). These populations were grown on Jiffy-7 peat pellets (Jiffy Products International, NL) under ambient (~420 ppm) or elevated (900 ppm) CO<sub>2</sub> in the growth chambers of the Microcosms experimental platform at the Montpellier European Ecotron CNRS. Five replicates of each accession were randomly distributed in the growth chambers. The concentration of 900 ppm of CO<sub>2</sub> was chosen as current CO<sub>2</sub> emissions align with the IPCC's RCP8.5 model, which predicts a CO<sub>2</sub> concentration of around 900 ppm in 2100. Growth conditions were 6-h/22-h light (22°C) / dark (20°C) photoperiod, with 200 µmol m<sup>-2</sup> s<sup>-1</sup> light intensity and 65% of hygrometry. Plants were watered twice a week with a growth solution containing KH<sub>2</sub>PO<sub>4</sub> 1 mM, MgSO<sub>4</sub> 1 mM, K<sub>2</sub>SO<sub>4</sub> 250 µM, CaCl<sub>2</sub> 250 µM, Na-Fe-EDTA 100 µM, KNO<sub>3</sub> 10 mM, KCl 50 µM, H<sub>3</sub>BO<sub>3</sub> 30 µM, MnSO<sub>4</sub> 5 µM, ZnSO<sub>4</sub> 1 µM, CuSO<sub>4</sub> 1 µM, (NH<sub>4</sub>)<sub>6</sub>Mo<sub>7</sub>O<sub>24</sub> 0.1 µM, as described by (Gansel et al., 2001 [↗](#)). The entire rosettes were collected three weeks after sowing. The *tip2;2-1* mutant line corresponds to the *SALK\_152463* allele (Wang et al., 2022 [↗](#)).

### Ionome analysis

From 3 to 5 replicates per accession were pooled and used for each ionome analysis. Total C and N content was obtained from dried shoot tissue using an Elementar Pyrocube analyzer. Cu, Fe, Mg, Mn, Na and Zn content was obtained from dry shoot tissue mixed with 750 µl of nitric acid (65% [v/v]) and 250 µl of hydrogen peroxide (30% [v/v]). After one night at room temperature, samples were mineralized at 85°C during 24 hours. Once mineralized, 4 ml of milliQ water was added to each sample. Mineral contents present in the samples were then measured by microwave plasma atomic emission spectroscopy (MP-AES, Agilent Technologies).

### Removal of outlier observations

Prior to GWAS and multivariate analyses such as PCA or clustering, mineral composition measures were pre-processed to remove technical outliers. For a given element and CO<sub>2</sub> condition, the values positioned more than 5 median absolute deviations away from the median were removed from the dataset. The number of outliers removed from each dataset is indicated in Supplemental Table 7.

## PCA and Clustering

Principal Component Analysis was performed using the R *ade4* package after the prior scaling of the variables to a z-score. Clustering of the REGMAP panel based on the relative changes of the mineral composition of each accession has been done using a k-means clustering with the R *kmeans* function. For this step, the variables were also scaled to a z-score. The number of clusters in the k-means algorithm was chosen by the elbow method on the criteria of cluster homogeneity (within-sum of squares).

## GWAs

Genome-Wide Association mapping was performed using the R *statgenGWAs* package. Genotype data was prepared using the *codeMarkers* function, removing duplicated SNPs and filtering for a minimum allele relative frequency of 0.04. Associations were performed by the *runSingleTraitGwas* function, that implements the EMMA algorithm. Population structure was modeled via a kinship matrix built from the Astle method. Manhattan plots were drawn using the *manPlotFast* function of the *ramwas* R package.

## RNA-seq experiments

Plants from the Columbia accession were grown in hydroponics to have access to the roots in addition to the shoot, as previously described in (Cassan et al., 2023 [DOI](#)). Shoot or root from 5 plants were pooled into one biological replicate, flash frozen in liquid nitrogen, and stored at -80°C. RNA of three biological replicates were extracted from shoot or root tissues using Direct-zol RNA Miniprep (Zymo Research, CA, USA), according to the manufacturer recommendations. RNA-sequencing libraries were done from shoot or root total RNA using standard RNA-Seq protocol method (Poly-A selection for mRNA species) by the Novogene company. RNA-sequencing was performed using Illumina technology on a NovaSeq6000 system providing PE150 reads. The quality control and adapter trimming of raw paired-end fastq files was done with *fastp* and its default parameters. Mapping to the TAIR10 reference genome was performed with STAR, and using the following options:

```
--outSAMtype BAM SortedByCoordinate
```

```
--outFilterMismatchNmax 1
```

```
--outFilterMismatchNoverLmax 0.15
```

```
--alignIntronMin 30
```

```
--alignIntronMax 5000
```

Quantification of the bam files against the TAIR10 GFF3 annotation file was done using *htseq-count* with options:

```
-f bam --type gene -r pos
```

```
--idattr=Name --stranded=no
```

Normalization and differential expression were performed using *DIANE* R package (Cassan et al., 2021 [DOI](#)), with no fold change constraint, and an adjusted p-value threshold (FDR) of 0.05. Lowly expressed genes with an average value across conditions under 25 reads were excluded from the analysis.

## Quantitative real-time PCR

Plants were grown in hydroponics to have access to the roots, as previously described in (Cassan *et al.*, 2023 [DOI](#)). Root tissue from 5 plants were pooled into one biological replicate, flash frozen in liquid nitrogen, and stored at -80°C. RNA were extracted from shoot or root tissues using TRIZOL (Invitrogen, USA), according to the manufacturer recommendations, and DNase treated using RQ1 (Promega, USA). Reverse transcription was achieved from 1 µg of total RNA with M-MLV reverse transcriptase (RNase H minus, Point Mutant, Promega, USA) using an anchored oligo(dT)20 primer. Accumulation of transcripts was measured by qRT-PCR (LightCycler 480, Roche Diagnostics, USA) using the SYBR Premix Ex Taq™ (TaKaRa, Japan). Gene expression was normalized using *UBQ10* and *ACT2* as internal standards. Results are presented as the expression relative to *UBQ10*. Sequences of primers used in RT-qPCR for gene expression analysis are listed in Supplemental Table 6.

## Acknowledgements

This work was supported by the I-Site Montpellier Université d'Excellence (MUSE; project ECO2THREATS), the CNRS through the Mission for Transversal and Interdisciplinary Initiatives (MITI) 80 PRIME program, and the Plant Biology and Breeding department of INRAE. O.C. was recipient of a PhD fellowship from the CNRS. T.M. was recipient of a PhD fellowship from INRAE and Région Occitanie. We thank Jiping Liu from Cornell University for the gift of *tip2;2-1* mutant line. This study benefited from the CNRS human and technical resources allocated to the Ecotron Research Infrastructure and from the state allocation “Investissement d'Avenir” AnaEE France ANR-11-INBS-0001. A CC-BY public copyright license has been applied by the authors to the present document.

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## Editors

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### Reviewer #1 (Public Review):

This study offers good evidence pointing to a genetic basis for *Arabidopsis thaliana*'s response to elevated CO<sub>2</sub> (eCO<sub>2</sub>) levels and its subsequent impact on the leaf ionome. The natural variation analyses in the study support the hypothesis that genetic factors, rather than local adaptation, guide the influence of eCO<sub>2</sub> on the ionome of rosette leaves in *Arabidopsis*.

Comments on current version:

I appreciate the revisions and the effort the authors have made.

Most of the abstract now accurately reflects the results and methods. It would be nice to have a few more technical details in the abstract, such as:

- \* What was the CO<sub>2</sub> level?
- \* Which gene was identified?

I still have a problem with this sentence:

"The elevation of atmospheric CO<sub>2</sub> leads to a decline in plant mineral content, which might pose a significant threat to food security in the coming decades."

The authors provide a wide range of published studies that support this statement. I fully agree that this is what the literature suggests. However, I think the literature has asked the wrong question.

In general, these studies addressed the question: Given no time for adaptation, do plants grown under high CO<sub>2</sub> have a different mineral composition? The answer is yes.

But a more important question is: Can plants and food crops adapt in time? I believe the strength of this study is that it tests this, and it suggests that the answer is yes. I also think there is a lot of unpublished results and greenhouse breeding success that supports the contention that most plants can adapt to the CO<sub>2</sub>.

"The artificial elevation of atmospheric CO<sub>2</sub> leads to a physiological response and decline in plant mineral content, which might pose a significant threat to food security in the coming decades if plants cannot adapt."

It needs to be made clear throughout the paper when high CO<sub>2</sub> levels lead to low mineral composition. These are all artificial manipulations without allowing the plants to adapt to the new environment.

"The elevation of atmospheric CO<sub>2</sub> concentration leads to a decline in the mineral composition of C3 plants (Gojon et al., 2023)." - this is well supported in artificial

environments.

Do wild plants have fewer minerals in their leaves today compared to plants in 1950? This would be great evidence and framing for this experiment.

Crop plants having lower nitrogen and different mineral compositions over time is substantially a product of breeders initially increasing inputs and then, over the last decade, selecting for higher input efficiency.

At the end of the introduction or the beginning of the results, please define why the CO<sub>2</sub> level was chosen and its context as being at the high end of current predictions.

"According to the literature, this results in a 20-25% reduction in vitamin C or lycopene and requires a significantly higher nitrogen and water intake to reach expected sugar levels (Doddrell H (2023), Horticulture Research). In addition, the negative effect of elevated CO<sub>2</sub> on tomato nutrient content seems to have significant repercussions on nutrition-health properties (Boufeldja (2023), Molecules)."

Thank you for sharing these reviews. These suggest to me that breeders favored the 80% yield bump over other traits. Either there was no breeding, or the breeding focused on other traits. It is important to mention that breeders should include mineral nutrition in their selection index while they maximize yield. Simpler breeding strategies can sometimes heavily favor one trait over others, but cattle breeders today regularly use selection indices that incorporate weights for two dozen traits.

This study provides nice evidence that an annual weed species is likely to be able to adapt easily to high eCO<sub>2</sub>. Whether perennial species will be able to adapt in time is clearly a topic that needs to be investigated.

<https://doi.org/10.7554/eLife.90170.2.sa1>

#### Reviewer #2 (Public Review):

The research uses a large collection of *Arabidopsis thaliana* accessions from various geographic scales to investigate the natural genetic variation underlying the response of ionome (elemental) composition to elevated CO<sub>2</sub> (eCO<sub>2</sub>), a concern for future food security. While most accessions show a decrease in elemental accumulation, the authors demonstrate a wide variety of responses to eCO<sub>2</sub> across the diversity of *Arabidopsis*, including lines that increase elemental content in eCO<sub>2</sub>. The demonstration of genetic diversity in eCO<sub>2</sub> response is a significant contribution to our understanding of this important phenomenon.

Comments on revised version:

The authors made significant improvements in the manuscript from the original preprint, and the conclusions are now well supported by the evidence presented.

<https://doi.org/10.7554/eLife.90170.2.sa0>

#### Author response:

The following is the authors' response to the original reviews.

##### ***eLife assessment***

*This paper provides useful information about how the ionome of *Arabidopsis thaliana* adapts to very high CO<sub>2</sub>-levels, backed up by solid evidence and carefully designed*

*studies. However, the broader claims of the paper about climate change and food security - heavily emphasized in the abstract, introduction, and discussion - are inappropriate, as there is no direct link to the presented work.*

We sincerely thank you for the work you have done in reviewing our manuscript. We very much appreciate your overall positive assessment of the experimental work as a whole, its value and robustness.

In this revised version, we took on board the majority of your suggestions and your comments. In particular, we understood your critical point about overstating our objectives, which might in turn seem uncorrelated with our results. We fully agree with the comments that have been made on this point. Consequently, we have made substantial modifications and corrections in order to clarify our objectives and their implications: exploring in depth the natural variation of the shoot ionome response to elevated CO<sub>2</sub>, and generating a valuable resource allowing a better understanding of the genetic and molecular mechanisms involved in the regulation of plant mineral nutrition by the elevation of atmospheric CO<sub>2</sub>.

We also made modifications in response to the other suggestions, including a clarification of the functional experiments carried out around the function of TIP2;2 in response to elevated CO<sub>2</sub>. Figure 7 now comprises the comparison between both ambient and elevated CO<sub>2</sub> conditions, which is much more informative than what appeared in the previous version.

#### **Public Reviews:**

##### **Reviewer #1 (Public Review):**

###### *Summary:*

*The study's abstract, introduction, and conclusions are not supported by the methods and results conducted. In fact, the results presented suggest that Arabidopsis could easily adapt to an extremely high CO<sub>2</sub> environment.*

We understand the reviewer's comment. Although our work is considered useful, robust and well designed, we agree with the reviewer's point. We have certainly overemphasized the significance of our work to address the issue of food security in response to rising atmospheric CO<sub>2</sub>, at the expense of the factual description of the results of our fundamental study of the mechanisms at the interface between CO<sub>2</sub> and mineral nutrition. We have clarified this focus by modifying the text of the introduction, objectives and discussion. We hope that these modifications will enable readers to better appreciate the core of this work.

Regarding the last part of the comment, our results do suggest that genetic variation could allow adaptation to rising atmospheric CO<sub>2</sub>, and our study does indeed aim to identify the extent and basis of this genetic variation.

*This study offers good evidence pointing to a genetic basis for Arabidopsis thaliana's response to elevated CO<sub>2</sub> (eCO<sub>2</sub>) levels and its subsequent impact on the leaf ionome. The natural variation analyses in the study support the hypothesis that genetic factors, rather than local adaptation, guide the influence of eCO<sub>2</sub> on the ionome of rosette leaves in Arabidopsis. However, the manuscript's claim regarding its role in "the development of biofortified crops adapted to a high-CO<sub>2</sub> world" (line 23) is overstated, especially given the absence of any analysis on the influence of eCO<sub>2</sub> on the seed ionome and Arabidopsis is a poor model for harvest index for any crop. The manuscript, in its current form, necessitates massive revisions, particularly in clarifying its broader implications and in providing more substantial evidence for some of its assertions.*

We thank the reviewer for this comment, and we would like to thank the reviewer for the positive appreciation for the identification of genetic basis for *Arabidopsis thaliana*'s response to elevated CO<sub>2</sub> and its subsequent impact on the leaf ionome. Nevertheless, it is true that the study of the leaf ionome is far from being able to lead to the development of biofortified plants. Some papers described that nutrient harvest index in *Arabidopsis* is a potential indicator of nutrient use efficiency (for instance, Masclaux-Daubresse and Chardon, Journal of Experimental Botany 2011 or Aranjuelo et al., Journal of Experimental Botany 2013). However, as we did not include any seed ionome data in the paper, we added clear mentions that our analyses were made on leaves (lines 56/57/250/319) and a comment in the discussion section to address this limitation (lines 325-328).

*Major Drawbacks and Questions:*

*(1) Evidence for the Central Premise:*

*The foundational premise of the study is the assertion that rising atmospheric CO<sub>2</sub> levels result in a decline in plant mineral content. This phenomenon is primarily observed in C<sub>3</sub> plants, with C<sub>4</sub> plants seemingly less affected. The evidence provided on this topic is scant and, in some instances, contradicts the authors' own references. The potential reduction of certain minerals, especially in grains, can be debated. For instance, reduced nitrogen (N) and phosphorus (P) content in grains might not necessarily be detrimental for human and animal consumption. In fact, it could potentially mitigate issues like nitrogen emissions and phosphorus leaching. Labeling this as a "major threat to food security" (line 30) is exaggerated. While the case for microelements might be more compelling, the introduction fails to articulate this adequately. Furthermore, the introduction lacks any discussion on how eCO<sub>2</sub> might influence nutrient allocation to grains, which would be crucial in substantiating the claim that eCO<sub>2</sub> poses a threat to food security. A more comprehensive introduction that clearly delineates the adverse effects of eCO<sub>2</sub> and its implications for food security would greatly enhance the manuscript.*

We partially agree with this comment. The decline in mineral status of C<sub>3</sub> plants under conditions of elevated atmospheric CO<sub>2</sub> has been widely described in the literature, and specifically documented for the cereal grains. While there are variations in this effect (depending on species, ecotype, cultivar), there is no debate about its acceptance. Here are just a few of the many works describing this effect, both on a global scale and at the level of the individual plant (Cotrufo MF (1998) Elevated CO<sub>2</sub> reduces the nitrogen concentration of plant tissues. Global Change Biology 4: 43-54; Loladze I (2014) Hidden shift of the ionome of plants exposed to elevated CO<sub>2</sub> depletes minerals at the base of human nutrition. eLife 3: e02245; Myers SS (2014) Increasing CO<sub>2</sub> threatens human nutrition. Nature 510: 139-142; Poorter H (1997) The effect of elevated CO<sub>2</sub> on the chemical composition and construction costs of leaves of 27 C<sub>3</sub> species. Plant, Cell & Environment 20: 472-482 ; Soares JC (2019) Preserving the nutritional quality of crop plants under a changing climate: importance and strategies. Plant and Soil 443: 1-26; Stitt M (1999) The interaction between elevated carbon dioxide and nitrogen nutrition: the physiological and molecular background. Plant, Cell & Environment 22: 583-621; Uddling J (2018) Crop quality under rising atmospheric CO<sub>2</sub>. Curr Opin Plant Biol 45: 262-267).

In addition to this, the threat to food security posed by this alteration in plant mineral status has also been well described in the literature by several modeling approaches (Beach RH (2019) Combining the effects of increased atmospheric carbon dioxide on protein, iron, and zinc availability and projected climate change on global diets: a modelling study. Lancet Planet Health 3: e307-e317; Ebi KL (2019) Elevated atmospheric CO<sub>2</sub> concentrations and climate change will affect our food's quality and quantity. Lancet Planet Health 3: e283-e284;

Medek DE (2017) Estimated Effects of Future Atmospheric CO<sub>2</sub> Concentrations on Protein Intake and the Risk of Protein Deficiency by Country and Region. *Environ Health Perspect* 125: 087002; Smith MR (2018) Impact of anthropogenic CO<sub>2</sub> emissions on global human nutrition. *Nature Climate Change* 8: 834-839; Weyant C (2018) Anticipated burden and mitigation of carbon-dioxide-induced nutritional deficiencies and related diseases: A simulation modeling study. *PLoS Med* 15: e1002586; Zhu C (2018) Carbon dioxide (CO<sub>2</sub>) levels this century will alter the protein, micronutrients, and vitamin content of rice grains with potential health consequences for the poorest rice-dependent countries. *Sci Adv* 4: eaaq1012). To reinforce this point, we have added a sentence and references (lines 30-33). Nevertheless, we understand the reviewer's comment on the nuance to be given to the intensity of this potential threat. We have therefore modified the text, replacing "major threat" by "significant threat" (lines 3 and 29).

We also would like to answer the reviewer's comment on the potential environmental benefit associated with reduced N and P content in grains (mitigation of N emissions and P leaching). Indeed, if this reduced N and P content results from a lowered use efficiency of soil nutrients by plants, as suggested by several studies (Bloom 2010, Cassan 2023, Gojon 2023 and references therein), this may at the opposite favor N oxides emission and P leaching from the soil.

#### (2) Exaggerated Concerns:

*The paper begins with the concern that carbon fertilization will lead to carbon dilution in our foods. While we indeed face numerous genuine threats in the coming decades, this particular issue is manageable. The increase in CO<sub>2</sub> alone offers many opportunities for boosting yield. However, the heightened heat and increased evapotranspiration will pose massive challenges in many environments.*

While there are indeed multiple threats that we are facing in the coming decades, we don't fully agree with this comment. At present, there's no evidence to say that the negative effect of CO<sub>2</sub> on plant mineral content will be manageable. Furthermore, there is compelling evidence that altered mineral nutrition and mineral status of plants will be an important factor limiting the high CO<sub>2</sub>-induced increase in yield, as will be heat or increased evapotranspiration (see for instance Coskun et al (2016) Nutrient constraints on terrestrial carbon fixation: The role of Nitrogen. *J. Plant Physiol.* 203: 95-109; Jiang M (2020) Low phosphorus supply constrains plant responses to elevated CO<sub>2</sub>: A meta-analysis. *Glob Chang Biol* 26: 5856-5873; Reich PB (2006) Nitrogen limitation constrains sustainability of ecosystem response to CO<sub>2</sub>. *Nature* 440: 922-925). Thus, although we do not negate the crucial importance of heat and water stress, we believe it is relevant to study the basic mechanisms responsible for the negative effect of CO<sub>2</sub> on plant mineral composition.

*Figure 4 in fact suggests that 43% of the REGMAP panel (cluster 3) is already pre-adapted to very high CO<sub>2</sub> levels. This suggests annual species could adapt very rapidly.*

We agree with the reviewer. However, this suggests that genetic variation exists in some ecotypes to support adaptation to elevated CO<sub>2</sub>. The purpose of this work is indeed to identify this genetic variation, in order to characterize the mechanisms behind.

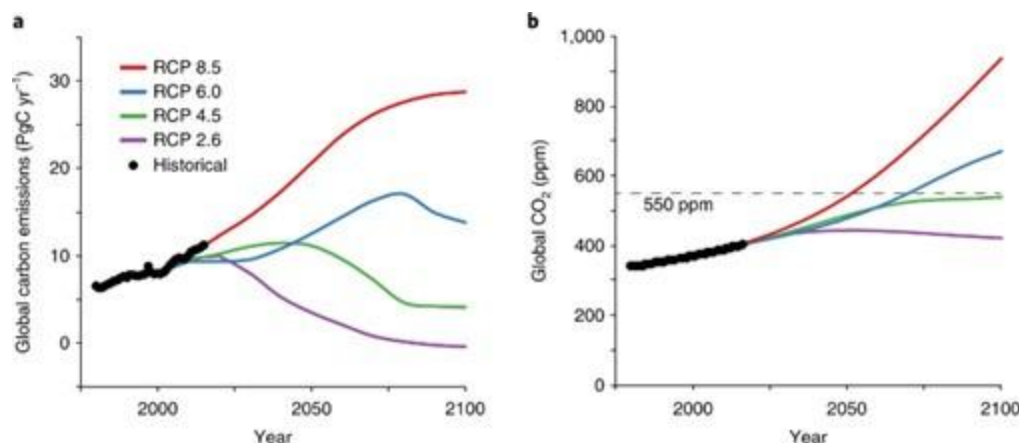
#### (3) Assumptions on CO<sub>2</sub> Levels:

*The assumption of 900ppm seems to be based on a very extreme climate change scenario. Most people believe we will overshoot the 1.5°C scenario, however, it seems plausible that 2.5 to 3°C scenarios are more likely. This would correspond to around 500ppm of CO<sub>2</sub>. <https://www.nature.com/articles/s41597-022-01196-7/tables/4>*



We agree with the reviewer that the CO<sub>2</sub> concentration we used corresponds to a high value in the IPCC projections. That said, this value is currently considered very plausible: the following figure (from Smith and Myers (2018) Nature Climate Change) shows that current CO<sub>2</sub> emissions align with the IPCC's most extreme model (RCP 8.5), which would result in a CO<sub>2</sub> concentration of around 900 ppm in 2100. Furthermore, nothing allows to exclude the 4°C scenario in the 6th IPCC report.

#### Author response image 1.



#### (4) Focus on Real Challenges:

*We have numerous real challenges, such as extreme heat and inconsistent rainfall, to address in the context of climate change. However, testing under extreme CO<sub>2</sub> conditions and then asserting that carbon dilution will negatively impact nutrition is exaggerated.*

While we fully agree that several threats linked to climate change exist, and all deserve to be studied, we find it questionable to consider that the potential effect of high CO<sub>2</sub> on the mineral nutrition of plants is not a real challenge. The mineral nutrition of plants is already a current major environmental challenge. This perspective seems to reflect the reviewer's personal opinion rather than an analysis of our work.

*In contrast, the FACE experiments are fundamental and are conducted at more realistic eCO<sub>2</sub> levels. Understanding the interaction between a 20% increase in CO<sub>2</sub> and new precipitation patterns is key for global carbon flux prediction.*

Again, we do not fully understand this comment, as the aim of our study was not to perform a global carbon flux prediction, but to unravel genes and mechanisms underlying the negative effect of elevated CO<sub>2</sub> on the nutrient content of *Arabidopsis* rosettes. However, we agree with the reviewer's comment and with the fact that FACE are useful facilities to explore the CO<sub>2</sub> response in more natural environments, and we highlight the fact that the decrease in mineral status of C<sub>3</sub> plants has been widely documented in FACE studies. FACE experiments do not facilitate, however, to conduct fully controlled experiments (temperature, rainfall, wind and light intensities are not controllable in FACE), that allow to disentangle the mechanisms by which elevated CO<sub>2</sub> regulates the signaling pathways associated with the plant mineral composition. In the longer term, studying the mechanisms we have identified in a more global context of climate change could be highly relevant.

*As I look at the literature on commercial greenhouse tomato production, 1000ppm of eCO<sub>2</sub> is common, but it also looks like the breeders and growers have already solved for*

*flavor and nutrition under these conditions.*

Indeed, tomato is often cultivated in CO<sub>2</sub>-enriched greenhouses at 1000 ppm. According to the literature, this results in a 20-25% reduction in vitamin C or lycopene, and requires a significantly higher nitrogen and water intake to reach expected sugar levels (Doddrell H (2023) Horticulture Research). In addition, the negative effect of elevated CO<sub>2</sub> on tomato nutrient content seems to have significant repercussions on nutrition-health properties (Boufeldja (2023), Molecules).

**Conclusion:**

*While the study provides valuable insights into the genetic underpinnings of Arabidopsis thaliana's response to elevated CO<sub>2</sub> levels, it requires an entirely revised writeup, especially in its abstract, broader claims and implications. The manuscript would benefit from a more thorough introduction, a clearer definition of its scope, and a clear focus on the limits of this study.*

We thank the reviewer for the comments made on our manuscript. In addition to the responses that we provide to these comments, we have modified the main text of the introduction, objectives and discussion to take these comments into consideration. We believe that this will significantly improve the manuscript.

**Reviewer #2 (Public Review):**

**Strengths:**

*The authors have conducted a large, well-designed experiment to test the response to eCO<sub>2</sub>. Overall, the experimental design is sound and appropriate for the questions about how a change in CO<sub>2</sub> affects the ionome of Arabidopsis. Most of the conclusions in this area are well supported by the data that the authors present.*

We thank the reviewer for this positive appreciation.

**Weakness:**

*While the authors have done good experiments, it is a big stretch from Arabidopsis grown in an arbitrary concentration of CO<sub>2</sub> to relevance to human and animal nutrition in future climates. Arabidopsis is a great model plant, but its leaves are not generally eaten by humans or animals.*

We agree with the reviewer's comment. We recognized that implying a direct contribution of our work to human nutrition in the future climates is overstated, as mentioned by the reviewer 1 as well. This was not an intentional overstatement, as we have always been convinced that our work contributed to the understanding of the basic mechanisms involved in the negative regulation of plant mineral nutrition by high CO<sub>2</sub>. We have significantly modified the text to correct any misunderstanding of our work's implication.

*The authors don't justify their choice of a CO<sub>2</sub> concentration. Given the importance of the parameter for the experiment, the rationale for selecting 900 ppm as elevated CO<sub>2</sub> compared to any other concentration should be addressed. And CO<sub>2</sub> is just one of the variables that plants will have to contend with in future climates, other variables will also affect elemental concentrations.*

We agree with this comment. We added a justification of the high CO<sub>2</sub> concentration used in this work in the Material and Methods section (lines 343-344). You can also read the

explanation of this choice in the response to the reviewer 1's point 3.

*Given these concerns, I think the emphasis on biofortification for future climates is unwarranted for this study.*

Anew, we agree with this comment and we have significantly modified the text to correct any misunderstanding of our work's implication.

*Additionally, I have trouble with these conclusions:*

*-Abstract "Finally, we demonstrate that manipulating the function of one of these genes can mitigate the negative effect of elevated CO<sub>2</sub> on the plant mineral composition."*

*-Discussion "Consistent with these results, we show that manipulating TIP2;2 expressions with a knock-out mutant can modulate the Zn loss observed under high CO<sub>2</sub>."*

*The authors have not included the data to support this conclusion as stated. They have shown that this mutant increases the Zn content of the leaves when compared to WT but have not demonstrated that this response is different than in ambient CO<sub>2</sub>. This is an important distinction: one way to ameliorate the reduction of nutrients due to eCO<sub>2</sub> is to try to identify genes that are involved in the mechanism of eCO<sub>2</sub>-induced reduction. Another way is to increase the concentration of nutrients so that the eCO<sub>2</sub>-induced reduction is not as important (i.e. a 10% reduction in Zn due to eCO<sub>2</sub> is not as important if you have increased the baseline Zn concentration by 20%). The authors identified tip2 as a target from the GWAS on difference, but their validation experiment only looks at eCO<sub>2</sub>.*

We thank the reviewer for this comment, and we agree with it. It is much more interesting, especially in the context of this paper, to analyze the function of a candidate gene not only in elevated CO<sub>2</sub>, but in both ambient and elevated CO<sub>2</sub>. Therefore, we added in Figure 7 data for the expression of TIP2;2 in contrasted haplotypes under ambient CO<sub>2</sub>, in comparison to those already presented under elevated CO<sub>2</sub> (now Fig. 7C and 7D). This showed that TIP2;2 expression is lower in haplotype 0 also under ambient CO<sub>2</sub>. We also added in Figure 7 (Fig. 7E) the Zn level in WT and tip2;2-1 mutant under ambient CO<sub>2</sub>, in comparison to those already presented under elevated CO<sub>2</sub>. This showed that the tip2;2-1 mutant line did not present any decrease in Zn shoot content in response to elevated CO<sub>2</sub>, in opposition to what is observed for the WT.

We have added comments associated to these new results in the Results and Discussion sections and in the discussion section (lines 233-242 in the results section, and lines 310-314 in the discussion section).

#### **Recommendations for the authors:**

##### **Reviewer #1 (Recommendations For The Authors):**

##### *Reviewer Comments on the Article's Approach to Ionome Analysis*

##### *(1) Omission of Phosphorus from the Ionome:*

*It's surprising that phosphorus (P) was not measured in the ionome. After nitrogen (N), P is often the most limiting mineral for plant development and yield, making it a significant component of the ionome. Why did the authors omit this crucial element?*

We agree with the reviewer that P is an important mineral for plant growth. The absence of data related to P content is due to feasibility constraints rather than oversight. The MP-AES instrument we used to analyze the ionome (except N and C, that we obtained from an

Elementar Analyzer) would have required an extra-step and an extra-analysis to obtain data for macronutrient such as P or K. In the context of this large-scale experiment, we faced the necessity to compromise and proceed without these data.

*(2) Relationship Between Leaf Ionome and Seed:*

*The manuscript lacks evidence demonstrating the relationship between the leaf ionome and the seed. This connection is vital to establish the study's aims as outlined in lines 20-24. If the central argument is that eCO<sub>2</sub> threatens food security, it's essential for the authors to either:*

- *Provide evidence that eCO<sub>2</sub> induces changes in the ionome profiles of seeds.*
- *Show that changes in the rosette leaf ionome lead to alterations in seed ionome profiles.*

We agree with the reviewer. Although we know that seed ionome composition of Arabidopsis model accession such as Columbia is indeed negatively affected by eCO<sub>2</sub>, we do not provide the data that support some of the terms used in lines 20-24. The correspondence between leaf and seed ionome in natural population under eCO<sub>2</sub> is certainly a next question that we will address. Therefore, to align our stated objectives with our data, we have modified the sentence in lines 20-24. We also added a comment on this point lines on the discussion section (lines 324-328).

*(3) Analysis of Ionome in Rosette Leaves:*

*Why did the authors choose to analyze the ionome specifically in rosette leaves? Is there a known correlation between the ionome profile in rosette leaves and seeds?*

See our answer to the above comment.

*(4) Experimental Design Comments:*

- *The layout of the accession growouts, the methods of randomization, blocking, and controls/checks should be detailed.*
- *Were BLUEs (Best Linear Unbiased Estimators) or BLUPs (Best Linear Unbiased Predictors) employed to account for experimental design conditions? If not, it's recommended that they be used.*

We thank the reviewer for this comment. A note on replicates has been added in the Method/Plant Material section. Concerning the BLUEs/BLUPs, although I am not familiar with their use, I do not think that these approaches are relevant in our experimental design. Indeed, we pooled 3 to 5 replicates for each accession to measure the ionome (as mentioned in the Method/Ionome analysis section – we realized this was perhaps not clear enough, and thus we reinforced this point in this section). Therefore, we do not have the variance data required to perform BLUEs/BLUPs.

*(5) Carbon Dilution Effect:*

*The statement, "The first component of the PCA described a clear antagonistic trend between C content and the change of other mineral elements (Fig. 3B)..." suggests a well-understood carbon dilution effect. These results are anticipated and align with existing knowledge.*

We thank the reviewer for this comment. However, this sentence does not relate to the biomass dilution hypothesis referred to by the reviewer. Indeed, the composition of each mineral (C and others) is expressed as a percentage of biomass, not as an absolute value. Therefore, this reflects more a probable effect of the increase in carbon compounds (notably soluble sugars), which could influence mineral composition.

*(6) Heritability Estimates:*

*The authors should report both the broad-sense heritability and an estimate of heritability based on a GRM or Kinship matrix.*

We thank the reviewer for this suggestion. We are skeptical of using a kinship matrix to estimate heritability in our study. Estimating narrow-sense heritability using a kinship matrix is conceptually based on the infinitesimal model of Fisher, thereby meaning that phenotypic variation is driven by hundreds to thousands of QTLs with small effects. If this is the case, GWAS conducted on several hundred (or even thousands) of genotypes will not be powerful enough to detect such QTLs. Accordingly, estimates of broad-sense heritability based on estimates of variance components can drastically differ from estimates of narrow-sense heritability based on the use of a kinship matrix, as illustrated in the study of Bergelson et al. (2019 Scientific Reports).

*(7) Application of the Breeder's Equation:*

*It would be beneficial if the authors applied the breeder's equation to estimate the species' potential rate of response. Based on the allele frequency of the adapted cluster 3 (69 ecotypes or 43% frequency of Figure 3B), it seems plausible that the populations could adapt within 23 generations.*

We thank the reviewer for this suggestion. Indeed, it would be really interesting to test whether sub-populations could adapt in comparison with others, and over what period of time. It is nevertheless not possible to do so using the Breeder's equation in our case, as this requires fitness data under conditions of ambient or elevated CO<sub>2</sub> (i.e. production of seeds) to be applied, and we do not have these data at the level of the whole population.

*(8) Overall Quality:*

In general, the authors have executed a high-quality ionome mapping experiment. However, the abstract, introduction, and discussion should be entirely rewritten and reframed.

We thank the reviewer for the positive evaluation of our experiment. As previously mentioned, we are for the most part in agreement with the comments made about the need to align our stated objectives with our experimental data and conclusions. To do so, we have rewritten part of the abstract, introduction and discussion. The details of these modifications are described in the responses made to each comment.

*Here's a line-by-line list of suggestions on writing:*

*Line 30 would read better with a comma after thus (or by replacing thus with therefore and then a comma at the start of the sentence).*

*Line 33 nevertheless would read better in between commas.*

*Lines 45 - 48 sentence is too long, could probably divide it into two.*

*Lines 90 - 94 are hard to interpret, recommend rephrasing for clarity.*

*Line 130 - keep verbs in the past tense for consistency (ran instead of run).*

*Line 194 - what do the authors mean by crossed? I'm inferring they looked at the intersection of DEGs with the list of genes identified by GWA mapping, probably should use a more concise word.*

*There's a concurrent use of the adjective strong (Lines 80, 142, 144, 197, 245). I would advise using a more concise adjective or avoiding its use to let the reader form their own opinion on the data.*

*Lines 174-176 the cited reference (No. 15) is incorrect. The study by Katz et al. (2022) does not provide information on the role of ZIF1 in zinc sequestration mechanisms under elevated CO2 conditions.*

We thank the reviewer for these detailed recommendations. We have corrected or rephrased the text according to these suggestions.

**Reviewer #2 (Recommendations For The Authors):**

*Technical points:*

*900 ppm as elevated CO2: Given the importance of the parameter for the experiment, the rationale for selection 900 ppm as elevated CO2 compared to any other concentration should be addressed.*

We acknowledge the reviewer's point and have previously addressed related aspects earlier in our response. In line with this, we have included a justification for this particular parameter in the Method section.

*The authors do not mention what genotype was used for their root/shoot RNAseq experiment.*

We thank the reviewer for this comment, and indeed, this information was not mentioned. This is now done, in the Method section.

*Line 125: Spelling error "REGMPA".*

This has been corrected.

*Line 338: Removal of outlier observations - "Prior to GWAS and multivariate analyses such as PCA or clustering, mineral composition measures were pre-processed to remove technical outliers". The authors should mention the exact number of outliers that were removed and what the explicit criteria were for removal.*

The number of outliers removed from each dataset is now indicated in Supplemental Table 7 (this is cited in the Method section). The explicit criteria used for this analysis is actually mentioned in the corresponding Method section: "the values positioned more than 5 median absolute deviations away from the median were removed from the dataset".

*Line 379: "Lowly expressed genes with an average value across conditions under 25 reads were excluded from the analysis". Providing information about the number of the lowly expressed genes that were removed from the analysis can help with the interpretation of the likelihood of the candidates selected being correct.*



This is a standard procedure in RNAseq analysis. It avoids many false positives in the differential analysis of gene expression based on ratios (where a very small number in the denominator can lead to a very high variation in expression, of no real significance). For information, this step led to the removal of 11607 and 10121 genes for the shoot and root datasets.

| *Line 384: It's not clear how many biological replicates were used.*

This has been corrected.

Additional comment: We have also become aware of a confusion concerning one of the candidate genes located close to GWA peaks: line 180 of the first version, we mentioned CAX1 (AT1G16380) for its role on nutrient deficiency response. There are actually two genes annotated as CAX1 in TAIR (both are cation exchangers), but the one involved in nutrient deficiency response is AT2G38170. We therefore removed the sentence mentioning AT1G16380/CAX1 as a potential candidate gene.

<https://doi.org/10.7554/eLife.90170.2.sa3>